Cyber Threat Detection, Monitoring, Alerting and Response Framework for Project PHOENIX

This document presents the design of detection, monitoring, alerting, and response mechanisms for identified cyber threats in Project PHOENIX. Based on the final threat dataset, each threat is analysed to define how it can be detected, what system activities should be monitored, when alerts should be triggered, and how mitigation and communication should be handled. This framework supports the development of a secure and responsive system for managing cyber risks during disaster scenarios.

Threat 1: Phishing (Fake Disaster Relief Email)

Field	Details
Detection	Suspicious links, urgent language, unknown sender domain, impersonation of trusted organizations
Activity to Track	Email logs, sender domain, URL click activity, message patterns
Alert	High
Mitigation	Block malicious links, quarantine email, prevent further delivery
Communication	Notify user about phishing attempt, alert security team

This attack involves emails impersonating disaster relief organizations to trick users into sharing sensitive information or making fraudulent payments. It exploits urgency and trust during disaster situations to increase success rates. Detection is performed by identifying suspicious links, unknown sender domains, and urgency-based language patterns in email content. The system also monitors user interaction with links and flags abnormal behavior. Once detected, the email is blocked and quarantined to prevent further impact. The user and security team are notified to ensure awareness and immediate response.

Threat 2: Ransomware Attack (Fire Rescue Victoria Case)

Field	Details
Detection	Unusual file encryption activity, unknown processes, sudden system changes
Activity to Track	File system logs, process activity, system performance metrics
Alert	Critical
Mitigation	Isolate affected system, stop malicious processes, prevent spread across network
Communication	Alert system administrator immediately, notify incident response team

This attack involves malicious software encrypting critical system data and demanding ransom for restoration. It can severely disrupt emergency services and compromise sensitive information. Detection is based on identifying abnormal file encryption behaviour, unexpected processes, and sudden system performance changes. Continuous monitoring of system logs helps in early identification of such threats. Once detected, the affected system is isolated to prevent further spread, and malicious processes are terminated. Immediate alerts are sent to administrators and the incident response team for rapid containment and recovery.

<u>Threat 3</u>: Scam (Fake Donation via Social Media)

Field	Details
Detection	Requests for donations, unverified accounts, emotional or urgent messaging
Activity to Track	Social media posts, user reports, transaction patterns
Alert	High
Mitigation	Flag content, block suspicious accounts, prevent transactions
Communication	Notify users, alert moderation and security team

This attack involves scammers using social media platforms to request fake donations during disasters. It exploits public sympathy and urgency to deceive users into sending money. Detection is based on identifying unverified sources, suspicious messaging patterns, and abnormal transaction behavior. Monitoring includes tracking user reports and social media activity. Once detected, the content is flagged, and accounts are restricted to prevent further spread. Users and moderation teams are notified to reduce impact and raise awareness.

<u>Threat 4</u>: API Abuse (Unsecured API Access)

Field	Details
Detection	Unusual API requests, high request frequency, unauthorized access patterns
Activity to Track	API logs, request frequency, authentication attempts
Alert	High
Mitigation	Revoke API access, enforce authentication, rate limiting
Communication	Notify system administrators and backend team

This attack occurs when attackers exploit unsecured APIs to access or manipulate sensitive data. It can lead to data breaches and service disruption. Detection is based on monitoring abnormal API usage, repeated requests, and unauthorized access attempts. API logs and authentication records help identify suspicious behaviour. Once detected, access is restricted and security controls such as authentication and rate limiting are enforced. Administrators are notified to take further action and secure the system.

<u>Threat 5:</u> IoT Attack (Sensor Manipulation)

Field	Details
Detection	Abnormal sensor readings, inconsistent data, unexpected device behaviour
Activity to Track	Sensor data logs, device IDs, communication patterns
Alert	High
Mitigation	Isolate affected device, ignore corrupted data
Communication	Notify system operator and technical team

This attack involves manipulation of IoT-based disaster monitoring sensors to produce false readings. It can lead to incorrect disaster predictions and delayed response. Detection is based on identifying abnormal sensor values and inconsistencies across devices. Monitoring includes analysing device logs and communication patterns. Once detected, the affected device is isolated and its data is ignored to prevent incorrect system decisions. Operators are notified to verify and restore normal functioning.

<u>Threat 6</u>: Data Breach (Unprotected API – Optus Case)

Field	Details
Detection	Unauthorized data access, abnormal data queries, exposure of sensitive information
Activity to Track	Access logs, API usage logs, database queries
Alert	High
Mitigation	Block access, secure API endpoints, enforce access controls
Communication	Notify administrators and affected stakeholders

This attack involves unauthorized access to sensitive data due to insecure APIs or system vulnerabilities. It can result in exposure of personal and confidential information. Detection is based on identifying abnormal access patterns and unexpected data queries. Monitoring includes reviewing API and database logs. Once detected, access is blocked and security controls are enforced to prevent further exposure. Administrators and affected users are notified to manage the incident.

<u>Threat 7</u>: DDoS Attack (Emergency Services Disruption)

Field	Details
Detection	Sudden traffic spikes, service slowdown, repeated requests from multiple sources
Activity to Track	Network traffic logs, request rates, IP activity
Alert	High
Mitigation	Rate limiting, block malicious IPs, scale resources
Communication	Notify administrators and network security team

This attack involves overwhelming system resources with excessive traffic, causing service disruption. It is especially critical during disaster situations where availability is essential. Detection is based on identifying unusual traffic spikes and repeated requests. Monitoring includes analysing network traffic and IP behaviour. Once detected, rate limiting and IP blocking are applied to control the attack. Administrators are alerted to ensure system stability and continued service availability.

<u>Threat 8</u>: Misinformation (Fake Disaster Updates)

Field	Details
Detection	Conflicting information, unverified sources, rapid spread of false content
Activity to Track	Social media feeds, content patterns, user reports
Alert	Medium

Mitigation	Flag content, require verification, restrict spread
Communication	Notify moderators and inform users

This attack involves spreading false information during disasters to create confusion and panic. It can reduce trust in official systems and delay response actions. Detection is based on identifying unverified sources and conflicting information patterns. Monitoring includes analysing social media content and user reports. Once detected, the content is flagged and restricted to prevent further spread. Moderators and users are informed to ensure awareness and verification.

<u>Threat 9</u>: Credential Theft (Unauthorized Access to Alert System)

Field	Details
Detection	Suspicious login activity, unknown locations, multiple failed attempts
Activity to Track	Login logs, IP addresses, authentication attempts
Alert	Critical
Mitigation	Lock account, enforce multi-factor authentication, revoke access
Communication	Notify administrators and affected users

This attack involves the use of stolen credentials to gain unauthorized access to critical systems such as alert platforms. It can lead to fake alerts and system misuse. Detection is based on monitoring unusual login behaviour and authentication failures. Tracking includes analysing login logs and IP patterns. Once detected, accounts are locked and additional authentication measures are enforced. Administrators and users are notified to secure access and prevent further compromise.

<u>Threat 10</u>: Deepfake / Misinformation (AI-Generated Content)

Field	Details
Detection	AI-generated content patterns, unverified media, inconsistencies in information
Activity to Track	Media content, source verification, social media spread
Alert	High
Mitigation	Flag content, verify authenticity, restrict distribution
Communication	Notify moderators and inform public users

This attack involves the use of AI-generated deepfake content to spread false disaster-related information. It can mislead the public and disrupt emergency response efforts. Detection is based on identifying inconsistencies in media and verifying source authenticity. Monitoring includes analysing content patterns and spread across platforms. Once detected, the content is flagged and restricted. Moderators and users are notified to ensure accurate information is shared.

<u>Threat 11</u>: Malware (Information Stealer)

Field	Details

Detection	Unusual background processes, data exfiltration behaviour, unknown applications
Activity to Track	Process logs, network activity, file access patterns
Alert	High
Mitigation	Remove malware, isolate system, block malicious processes
Communication	Notify security team and system administrator

This attack involves malware that steals user credentials and sensitive system information. It operates silently in the background, making detection difficult. Detection is based on identifying abnormal process behaviour and unusual network communication. Monitoring includes tracking file access and outbound data transfers. Once detected, the system is isolated and malicious processes are removed. Security teams are notified to prevent further compromise.

<u>Threat 12</u>: Voice Phishing (Vishing Attack)

Field	Details
Detection	Suspicious calls requesting sensitive data, unusual remote access requests
Activity to Track	Call logs, user activity, remote access attempts
Alert	High
Mitigation	Block access, educate users, restrict unauthorized tools
Communication	Notify users and security awareness team

This attack involves phone-based social engineering where attackers trick users into installing malicious tools or revealing credentials. It relies heavily on trust and urgency. Detection is based on identifying unusual remote access requests and suspicious communication patterns. Monitoring includes tracking user activity and system changes after such interactions. Once detected, access is blocked and users are alerted. Security teams take action to prevent further incidents.

<u>Threat 13</u>: Fraudulent Applications (Disaster Relief Fraud)

Field	Details
Detection	Duplicate submissions, fake data patterns, automated entries
Activity to Track	Application logs, user submissions, IP patterns
Alert	Medium
Mitigation	Reject fraudulent entries, apply validation checks
Communication	Notify administrators and verification team

This attack involves submitting fake disaster relief applications to exploit financial aid systems. It can cause financial loss and delay genuine claims. Detection is based on identifying duplicate or suspicious application patterns. Monitoring includes tracking submission behaviour and IP activity. Once detected, fraudulent entries are rejected and validation mechanisms are strengthened. Administrators are informed for further verification.

<u>Threat 14</u>: Fake Charity Scam (Flood/Bushfire Context)

Field	Details
Detection	Unverified charity sources, suspicious payment requests
Activity to Track	Transaction logs, website activity, user reports
Alert	High
Mitigation	Block fraudulent accounts, flag transactions
Communication	Notify users and fraud prevention team

This attack involves fake charity organisations collecting donations during disasters. It exploits public generosity and urgency. Detection is based on identifying unverified sources and abnormal payment patterns. Monitoring includes tracking transactions and user complaints. Once detected, accounts are blocked and transactions are flagged. Users are warned to avoid further financial loss.

***Threat 15*: Cyber Attack on Infrastructure (DP World Case)**

Field	Details
Detection	System disruptions, unusual access patterns, service outages
Activity to Track	Network logs, system uptime, access logs
Alert	Critical
Mitigation	Isolate affected systems, restore services, block attackers
Communication	Notify administrators and emergency response teams

This attack targets critical infrastructure systems causing major service disruptions. It can affect supply chains and essential operations. Detection is based on identifying system failures and abnormal access behaviour. Monitoring includes analysing network and system logs. Once detected, systems are isolated and recovery procedures are initiated. Authorities and response teams are informed immediately.

Threat 16: Unauthorized Access (Stolen Credentials – Alert System)

Field	Details
Detection	Login from unknown locations, unusual system actions
Activity to Track	Login logs, access behaviour, session activity
Alert	Critical
Mitigation	Revoke access, enforce MFA, reset credentials
Communication	Notify administrators and affected users

This attack involves unauthorized access to systems using stolen credentials. It can result in misuse of critical functions like sending fake alerts. Detection is based on identifying abnormal login behaviour and unusual system activity. Monitoring includes tracking session and access logs. Once detected, access is revoked and credentials are reset. Administrators and users are notified immediately.

Threat 17: Human Error (Misconfiguration)

Field	Details
Detection	Unexpected system behaviour, misconfigured settings
Activity to Track	Configuration logs, system changes
Alert	Medium

Mitigation	Correct configurations, apply security controls
Communication	Notify administrators and technical team

This issue occurs when system configurations are incorrectly set, leading to vulnerabilities. It is unintentional but can have serious impacts. Detection is based on identifying abnormal system behaviour and configuration changes. Monitoring includes tracking system settings and updates. Once detected, configurations are corrected, and controls are applied. Technical teams are informed for validation.

<u>Threat 18</u>: Social Engineering (Door-to-door Scam)

Field	Details
Detection	Reports of suspicious activity, unusual requests for payments
Activity to Track	User reports, complaint logs
Alert	Medium
Mitigation	Warn users, verify identities
Communication	Notify community and awareness teams

This attack involves individuals physically approaching victims to collect money under false pretences. It exploits trust and lack of verification. Detection is based on user reports and unusual activity patterns. Monitoring includes analysing complaints and reports. Once detected, users are warned and verification processes are enforced. Awareness teams are notified to prevent further incidents.

<u>Threat 19</u>: Ransomware (Healthcare System Target)

Field	Details
Detection	System lockout, encrypted data, ransom messages
Activity to Track	System logs, file changes, process activity
Alert	Critical
Mitigation	Isolate system, backup recovery, remove malware
Communication	Notify administrators and incident response team

This attack targets healthcare systems, encrypting data and demanding ransom. It can disrupt critical services and expose sensitive information. Detection is based on identifying encrypted files and ransom indicators. Monitoring includes tracking system activity and file changes. Once detected, systems are isolated and recovery processes begin. Response teams are alerted immediately.

<u>Threat 20</u>: Unprotected API (Optus Data Breach)

Field	Details
Detection	Unauthorized API access, abnormal data queries
Activity to Track	API logs, database queries, access records
Alert	High

Mitigation	Secure API endpoints, enforce authentication
Communication	Notify administrators and stakeholders

This attack involves exploitation of unsecured APIs leading to data exposure. It can affect millions of users. Detection is based on identifying unusual API usage and unauthorized access. Monitoring includes analysing API and database logs. Once detected, APIs are secured and access controls are enforced. Stakeholders are informed to manage the breach.

<u>Threat 21</u>: Data Theft (Latitude Financial Case)

Field	Details
Detection	Unusual data access, large data transfers
Activity to Track	Access logs, network traffic, database queries
Alert	High
Mitigation	Restrict access, monitor activity, secure systems
Communication	Notify administrators and affected users

This attack involves unauthorized access to sensitive customer data through compromised systems. It can lead to identity theft and financial loss. Detection is based on monitoring abnormal data access patterns. Monitoring includes analysing network traffic and database logs. Once detected, access is restricted and systems are secured. Users and administrators are informed.

<u>Threat 22</u>: Phishing / Scam (Government Impersonation SMS)

Field	Details
Detection	Fake SMS messages, malicious links, impersonation
Activity to Track	SMS logs, link clicks, user reports
Alert	High
Mitigation	Block links, flag messages
Communication	Notify users and security team

This attack involves SMS messages impersonating government authorities to steal user data. It exploits trust in official communication. Detection is based on identifying suspicious messages and links. Monitoring includes tracking SMS activity and user interactions. Once detected, messages are blocked and users are warned. Security teams are informed for further action.

<u>Threat 23</u>: Fraudulent Claims (Government Payment Fraud)

Field	Details
Detection	Fake identities, duplicate claims, abnormal submissions
Activity to Track	Claim logs, identity verification, IP activity
Alert	Medium
Mitigation	Reject claims, apply verification checks
Communication	Notify administrators and fraud team

This attack involves submitting fake claims to receive disaster relief payments. It can cause financial losses and misuse resources. Detection is based on identifying duplicate or suspicious claim patterns. Monitoring includes analysing submission behaviour. Once detected, claims are rejected and verification processes are enforced. Administrators are notified.

<u>Threat 24</u>: Scam (Flood-related Multi-channel Attack)

Field	Details
Detection	Multiple communication channels, suspicious links, impersonation
Activity to Track	Email, SMS, calls, user reports
Alert	High
Mitigation	Block communication, warn users
Communication	Notify users and security team

This attack uses multiple communication methods to target victims during floods. It increases reach and effectiveness. Detection is based on identifying patterns across different channels. Monitoring includes analysing communication logs and reports. Once detected, communication is blocked and users are warned. Security teams take action.

<u>Threat 25</u>: Fraudulent Grant Claims (Flood Recovery)

Field	Details
Detection	Fake documents, repeated submissions, abnormal patterns
Activity to Track	Submission logs, document verification
Alert	Medium
Mitigation	Reject claims, verify documents
Communication	Notify administrators

This attack involves submitting fake documents to claim financial assistance. It exploits disaster recovery systems. Detection is based on identifying fake or repeated submissions. Monitoring includes verifying documents and submission logs. Once detected, claims are rejected. Administrators are informed for further review.

<u>Threat 26</u>: Data Breach (Sensitive Data Exposure)

Field	Details
Detection	Unauthorized data exposure, abnormal access
Activity to Track	Access logs, database activity
Alert	High
Mitigation	Secure data, restrict access
Communication	Notify administrators and affected users

This attack involves exposure of sensitive personal data due to poor security practices. It can lead to privacy violations and identity theft. Detection is based on identifying abnormal access patterns. Monitoring includes analysing data access logs. Once detected, access is restricted and data is secured. Affected users are informed.

<u>Threat 27</u>: General Phishing Attack

Field	Details
Detection	Suspicious emails, fake links, impersonation
Activity to Track	Email logs, user activity
Alert	High
Mitigation	Block messages, quarantine emails
Communication	Notify users and security team

This attack involves sending phishing messages to trick users into revealing sensitive data. It is common and widespread. Detection is based on identifying suspicious email patterns. Monitoring includes tracking email activity. Once detected, messages are blocked and users are warned. Security teams are informed.

<u>Threat 28</u>: Social Engineering (Fake Repair Services)

Field	Details
Detection	Unverified service requests, suspicious payment demands
Activity to Track	User reports, payment logs
Alert	Medium
Mitigation	Verify service providers, warn users
Communication	Notify community and support teams

This attack involves scammers posing as repair workers targeting disaster victims. It exploits urgency and lack of verification. Detection is based on identifying suspicious service requests. Monitoring includes analysing reports and payment patterns. Once detected, users are warned and verification is enforced. Support teams are notified.

<u>Threat 29</u>: Misinformation (Bushfire Context)

Field	Details
Detection	False information, unverified sources, conflicting data
Activity to Track	Social media, content patterns
Alert	Medium
Mitigation	Flag content, verify information
Communication	Notify moderators and users

This attack involves spreading false information during bushfires to create confusion. It can impact public safety. Detection is based on identifying misinformation patterns. Monitoring includes analysing social media content. Once detected, content is flagged and verified. Users are informed to avoid confusion.

<u>Threat 30</u>: Scam (Fake Flood Relief Payment Messages)

Field	Details
Detection	Fake payment messages, suspicious links, impersonation
Activity to Track	SMS logs, user reports

Alert	High
Mitigation	Block messages, warn users
Communication	Notify users and security team

This attack involves fake messages offering disaster relief payments to trick victims into sharing personal information. It exploits urgency and trust. Detection is based on identifying suspicious messages and links. Monitoring includes tracking SMS activity and reports. Once detected, messages are blocked and users are warned. Security teams are informed.

<u>Threat 31</u>: Data Breach (Medibank Case)

Field	Details
Detection	Unusual database access, large data extraction, unauthorized queries
Activity to Track	Database logs, access records, network traffic
Alert	Critical
Mitigation	Restrict access, isolate systems, secure databases
Communication	Notify administrators, affected users, and response team

This attack involves unauthorized access and extraction of sensitive customer data from healthcare systems. It can lead to identity theft and privacy violations. Detection is based on identifying abnormal database queries and large-scale data access. Monitoring includes analysing database and network logs. Once detected, access is restricted and systems are secured. Administrators and affected users are notified immediately.

<u>Threat 32</u>: Data Breach (Optus Case)

Field	Details
Detection	Unauthorized access, abnormal API usage, large data queries
Activity to Track	API logs, database activity, access patterns
Alert	High
Mitigation	Secure APIs, enforce authentication, restrict access
Communication	Notify administrators and affected users

This attack involves exposure of customer data due to insecure API endpoints. It affects a large number of users and compromises sensitive information. Detection is based on identifying abnormal API requests and unauthorized access. Monitoring includes analysing API logs and database queries. Once detected, APIs are secured and access is restricted. Administrators and users are informed to manage the breach.

<u>Threat 33</u>: Infrastructure Disruption (Port Cyber Attack)

Field	Details
Detection	System outages, abnormal network behaviour, service disruptions
Activity to Track	Network logs, system uptime, service availability
Alert	Critical
Mitigation	Isolate affected systems, restore services, block attack sources
Communication	Notify administrators and emergency response teams

This attack targets infrastructure systems causing service disruptions and operational delays. It can affect supply chains and essential services. Detection is based on identifying system downtime and abnormal network activity. Monitoring includes analysing system performance and network logs. Once detected, systems are isolated and recovery processes are initiated. Relevant teams are informed immediately.

<u>Threat 34</u>: Data Breach (Latitude Financial Case)

Field	Details
Detection	Unauthorized data access, large data transfers, abnormal behaviour
Activity to Track	Database logs, network traffic, user activity
Alert	High
Mitigation	Restrict access, secure systems, monitor activity
Communication	Notify administrators and affected users

This attack involves unauthorized access to financial data through compromised systems. It can lead to identity theft and financial loss. Detection is based on identifying unusual data access patterns. Monitoring includes analysing logs and network activity. Once detected, access is restricted and systems are secured. Users and administrators are informed.

<u>Threat 35</u>: Emergency Alert Compromise

Field	Details
Detection	Unauthorized alert generation, unusual alert patterns
Activity to Track	Alert logs, system access logs
Alert	Critical
Mitigation	Disable alert system, verify alerts, revoke access
Communication	Notify authorities and administrators immediately

This attack involves sending fake emergency alerts using compromised systems. It can create panic and reduce trust in official communications. Detection is based on identifying unauthorized alert activity and abnormal patterns. Monitoring includes analysing alert system logs. Once detected, the alert system is secured and false alerts are blocked. Authorities are notified immediately.

<u>Threat 36</u>: Charity Scam (Bushfire Context)

Field	Details
Detection	Fake charity requests, suspicious payment links
Activity to Track	Transaction logs, website activity, user reports
Alert	High
Mitigation	Block fraudulent accounts, flag transactions
Communication	Notify users and fraud prevention team

This attack involves scammers impersonating charities to collect donations during disasters. It exploits public trust and urgency. Detection is based on identifying suspicious payment requests

and unverified sources. Monitoring includes tracking transactions and user reports. Once detected, accounts are blocked and users are warned. Fraud teams are notified.

<u>Threat 37</u>: Donation Fraud (Fake Campaigns)

Field	Details
Detection	Fake fundraising campaigns, duplicate requests
Activity to Track	Campaign data, user reports, transaction logs
Alert	High
Mitigation	Remove fraudulent campaigns, block accounts
Communication	Notify users and platform administrators

This attack involves fake fundraising campaigns targeting disaster victims. It can lead to financial loss and misuse of funds. Detection is based on identifying duplicate or suspicious campaigns. Monitoring includes analysing campaign data and reports. Once detected, campaigns are removed and accounts are blocked. Users are informed.

<u>Threat 38</u>: Malware (Information Stealer Campaign)

Field	Details
Detection	Suspicious applications, data exfiltration behaviour
Activity to Track	Process logs, network traffic, file access
Alert	High
Mitigation	Remove malware, isolate system
Communication	Notify security team

This attack involves malware designed to steal user credentials and system data. It operates stealthily and can cause major data breaches. Detection is based on identifying abnormal processes and network activity. Monitoring includes analysing system logs. Once detected, the system is isolated and malware is removed. Security teams are informed.

<u>Threat 39</u>: Network Intrusion (Port Systems)

Field	Details
Detection	Unauthorized network access, unusual traffic patterns
Activity to Track	Network logs, access records
Alert	High
Mitigation	Block access, isolate network segments
Communication	Notify administrators and network team

This attack involves unauthorized access to network systems, leading to disruptions. It can impact critical infrastructure. Detection is based on identifying abnormal network behaviour. Monitoring includes analysing network logs. Once detected, access is blocked and networks are secured. Administrators are informed.

<u>Threat 40</u>: Misinformation Campaign (Bushfire Context)

Field	Details
Detection	False narratives, rapid spread, unverified sources
Activity to Track	Social media feeds, content analysis
Alert	Medium
Mitigation	Flag content, verify sources
Communication	Notify moderators and users

This attack involves coordinated spread of misinformation during disasters. It can create confusion and reduce trust. Detection is based on identifying false content patterns. Monitoring includes analysing social media activity. Once detected, content is flagged and verified. Users are informed.

<u>Threat 41</u>: Phishing (Government Impersonation Email)

Field	Details
Detection	Fake government emails, malicious links
Activity to Track	Email logs, user activity
Alert	High
Mitigation	Block emails, quarantine messages
Communication	Notify users and security team

This attack involves phishing emails impersonating government agencies to steal user data. It exploits trust in official communication. Detection is based on identifying suspicious email patterns. Monitoring includes tracking user interactions. Once detected, emails are blocked and users are warned. Security teams are informed.

<u>Threat 42</u>: Ransomware (Healthcare Organisations)

Field	Details
Detection	Encrypted systems, ransom messages
Activity to Track	System logs, file changes
Alert	Critical
Mitigation	Isolate system, recover from backups
Communication	Notify administrators and response team

This attack targets healthcare systems with ransomware, disrupting services. It can expose sensitive data. Detection is based on identifying encrypted files and ransom indicators. Monitoring includes analysing system logs. Once detected, systems are isolated and recovery begins. Teams are notified immediately.

<u>Threat 43</u>: Phishing / Social Engineering (Fake Emergency SMS)

Field	Details
Detection	Fake SMS alerts, malicious links
Activity to Track	SMS logs, user reports

Alert	High
Mitigation	Block messages, warn users
Communication	Notify users and security team

This attack involves fake emergency SMS messages to deceive users. It can cause panic and data theft. Detection is based on identifying suspicious messages. Monitoring includes analysing SMS activity. Once detected, messages are blocked and users are warned. Security teams are informed.

<u>Threat 44</u>: AI-generated Phishing (Relief Organisation Impersonation)

Field	Details
Detection	AI-generated content patterns, suspicious emails
Activity to Track	Email logs, content analysis
Alert	High
Mitigation	Block emails, flag content
Communication	Notify users and moderators

This attack uses AI-generated phishing emails to deceive users. It increases scale and sophistication. Detection is based on identifying unusual content patterns. Monitoring includes analysing email data. Once detected, emails are blocked and flagged. Users are informed.

<u>Threat 45</u>: Unauthorized Access (Alert System Misuse)

Field	Details
Detection	Suspicious access, abnormal system actions
Activity to Track	Access logs, system activity
Alert	Critical
Mitigation	Revoke access, secure system
Communication	Notify administrators immediately

This attack involves misuse of alert systems through unauthorized access. It can disrupt emergency communication. Detection is based on identifying abnormal system behaviour. Monitoring includes analysing logs. Once detected, access is revoked and systems are secured. Administrators are notified.

<u>Threat 46</u>: Malware (Fake Mobile Apps)

Field	Details
Detection	Suspicious apps, abnormal device behaviour
Activity to Track	App activity, device logs
Alert	High
Mitigation	Remove apps, secure devices
Communication	Notify users and security team

This attack involves malicious apps impersonating disaster services to steal user data. It affects mobile users. Detection is based on identifying suspicious app behaviour. Monitoring includes analysing device logs. Once detected, apps are removed and users are warned. Security teams are informed.

<u>Threat 47</u>: Misinformation (Fake Evacuation Updates)

Field	Details
Detection	False updates, unverified sources
Activity to Track	Social media, user reports
Alert	Medium
Mitigation	Flag content, verify information
Communication	Notify moderators and users

This attack spreads false evacuation information during disasters. It can risk public safety. Detection is based on identifying unverified updates. Monitoring includes analysing social media content. Once detected, content is flagged and verified. Users are informed.

<u>Threat 48</u>: DDoS Attack (Emergency Websites)

Field	Details
Detection	Traffic spikes, service disruption
Activity to Track	Network logs, traffic patterns
Alert	High
Mitigation	Rate limit, block IPs
Communication	Notify administrators

This attack targets emergency response websites with heavy traffic to disrupt access. It affects availability. Detection is based on identifying traffic spikes. Monitoring includes analysing network logs. Once detected, traffic is controlled and systems are stabilised. Administrators are informed.

<u>Threat 49</u>: Data Breach (Disaster Victim Data Exposure)

Field	Details
Detection	Unauthorized data access, exposure of sensitive records
Activity to Track	Database logs, access activity
Alert	High
Mitigation	Secure data, restrict access
Communication	Notify administrators and affected users

This attack involves exposure of sensitive disaster victim data due to poor security. It can lead to privacy risks. Detection is based on identifying abnormal access. Monitoring includes analysing logs. Once detected, data is secured and access is restricted. Users are informed.

<u>Threat 50</u>: Phishing Website (Fake Government Portal)

Field	Details
Detection	Fake websites, credential harvesting attempts
Activity to Track	Web traffic, user reports
Alert	High
Mitigation	Block website, warn users
Communication	Notify users and security team

This attack involves fake websites mimicking official disaster relief portals to steal credentials. It exploits trust in government systems. Detection is based on identifying suspicious websites. Monitoring includes analysing traffic and reports. Once detected, websites are blocked and users are warned. Security teams are informed.